Dear Editors of the Journal of Machine Learning for Biomedical Imaging,

We submit for your consideration "When Leakage Changes the Conclusion: A Methodological Evaluation of Segmentation-Guided Burn Severity Classification."

A burn photograph contains the wound and a great deal besides: unburned skin, dressings, bedding, the room. A classifier trained on whole images is free to read the setting rather than the injury, and in a web-sourced collection the setting correlates with severity for reasons unrelated to tissue. The obvious remedy is to segment the burn and grade only what remains. That design is standard elsewhere in medical imaging — dual-stage frameworks localise a region of interest before classifying it — but for burns it had not been evaluated against the simpler alternative it is meant to improve on.

We built the system to find out: a YOLOv8x-seg localiser feeding a masked Swin classifier, served through a Flutter and Flask mobile application. We then built a benchmark to test it honestly, and that honest test is the substance of the paper. Three things may interest your readership.

1. A leakage artifact whose structure differs from the published cases. Our original result was that masking the image to the burn improved accuracy by 3.06 percentage points across 20 of 21 architectures. It was an artifact: the dataset augments each photograph into about 2.5 copies with distinct file names, and a file-level split placed copies of one photograph in both folds. In the masked classification set, 80.5 percent of test images shared a source photograph with training; the separately partitioned unmasked set leaked none. Source-grouped, the effect is +0.55 points (95 percent CI −0.37 to +1.47). Restricted to the eight architectures common to both analyses the matched contrast is 2.07 → 0.55 rather than 3.06 → 0.55, and we rely on the smaller figure.

What we believe is unusual here is not the magnitude, which is smaller than the published quantified-leakage studies, but that **the leak rate was confounded with the condition under comparison** — 80.5 percent in one arm, zero in the other. Leakage therefore did not inflate a score; it manufactured a comparative conclusion and a design recommendation, consistently across 20 of 21 architectures. The nearest precedent we found is confound-leakage (Hamdan et al., 2023), and we scope our claim against it explicitly rather than claiming priority.

2. Direct evidence that a three-seed protocol is actively misleading, not merely imprecise. Three seeds put the internal head-to-head margin at +3.74 points, 95 percent CI [+3.04, +4.44], p = 0.002 — an interval about three times narrower than the data support — around an effect whose true magnitude is smaller. The same three seeds put the external margin at +3.24, CI [−3.83, +10.31], p = 0.19, licensing no claim at all, when that effect is in fact the most reliable in the study (ten seeds: +2.79, CI [+1.09, +4.49], p = 0.005, ahead in 9 of 10 runs). One protocol overstated the weaker result and concealed the stronger one simultaneously.

Because that comparison also involved a recipe change, we removed the confound: across all 120 three-seed subsets **of the same ten runs**, only 15 detect the external effect, estimates range from +0.21 to +5.22 externally and −0.98 to +6.67 internally — some subsets place the pipeline ahead — and the estimated standard deviation ranges from 0.00 to 3.25 percentage points.

3. External validation that fails, reported as such. On the independent BIP_US clinical database the system assigns the mildest grade to 48 of 94 images — a class that database does not contain — so its below-chance balanced accuracy measures a systematic severity offset rather than an inability to rank depth. The direction is the harmful one either way. On a second external set of 319 images, screened by perceptual hashing that removed 118 training-identical images, it under-graded 28.3 percent of the images that could be under-graded against 19.3 percent internally. Because those 319 images come from 175 source photographs, we report source-clustered bootstrap intervals alongside the Wilson intervals; they are about a fifth wider, and they are the ones we rely on.

4. A fairness probe that we could have published as a positive finding. An exploratory skin-tone analysis gave a large internal subgroup effect — median ITA 19.2 for errors against 38.4 for correct predictions, p = 0.0025, surviving Bonferroni correction across all 15 tests, in the direction the fairness literature would predict. It does not replicate: the same test on the external set, better powered at 139 images against 81, gives p = 0.35 with the effect reversed. We report it in full, as a null with the subgroup signal disclosed, because deleting it would be exactly the selective reporting the paper argues against.

5. A leak we found in our own pipeline, and did not quietly fix. While packaging the code for release we discovered that our source-grouped split had been built for the classifiers and never propagated to the segmentation models, which kept the collection's original split. As a result 179 of the 205 internal test images — 87.3 percent — sit in the segmentation models' training or validation folds. We could have retrained quietly and said nothing. Instead we quantified it: the standalone segmentation model scored 90.7 percent internally against 48.9 externally, a 41.8-point gap that is the signature of being scored on one's own training data. Then we measured the leak by removing it: we rebuilt both segmentation datasets on the classifiers' source-grouped split and retrained both models. We had described that retraining as changing nothing but the partition; reading the original checkpoints' embedded training arguments during the final audit showed otherwise, and the manuscript now says so. The originals ran at seed 0 with patience 50, and the localiser used a learning rate of 10^{-4} and was continued from an earlier checkpoint rather than the public weights, so the retraining differs in seed, patience, and for the localiser in learning rate and initialisation as well as in the split. On the same 205 images the standalone model falls from 90.7 to **78.0 percent**. We then retrained the pipeline's ten Swin-Tiny classifiers and re-scored that arm on the corrected localiser too — and found the effect negligible: 80.98 to 80.88 percent. A leak worth 12.7 points to one model was worth 0.10 to another, because in the pipeline the grading is done by a stage that was never contaminated. Contamination does not propagate uniformly through a pipeline. We report all of it, including our own wrong expectation, because a paper arguing that leakage is easy to introduce and hard to see should say plainly when it introduced some and could not see it.

6. The metric decides the internal result. The internal margin holds on plain accuracy (+2.73, p = 0.016) but not on balanced accuracy (+1.65, p = 0.10). Only the external margin survives both summaries.

7. We selected our own headline arm on the test set. ConvNeXt-Large was chosen as the best of eleven Benchmark 1 architectures — best *on test*. On validation Swin-Tiny wins, and scores 2.4 points lower on test. Against the seed-42 pipeline the margin falls from +3.41 to +0.98: honest selection removes **71 percent** of the internal margin. We therefore report no internal effect estimate at all. This is the audit that costs us most, and it is in the paper because it is true.

8. We then did the same thing again, and found it only at the very end. The paper's end-to-end conclusion rested on the pipeline's "best configuration", regime B of three. Regime B is the best **on test**. Ranked on validation, which is the honest basis, the order reverses: regime C leads at 84.37 percent against B's 83.45 and A's 83.35. That matters, because against the validation-selected regime C the external difference from the plain classifier **is** resolved — 2.57 percentage points, 95 percent interval +0.87 to +4.27, $p = 0.008$ — where against test-selected regime B it is not ($p = 0.11$). An earlier version of this manuscript therefore claimed the two systems were statistically indistinguishable on **both** test sets. That claim held only under a configuration we had chosen on the sets we then scored it on. We have withdrawn the external half of it. Internal parity survives either choice ($p = 0.65$ for B, 0.67 for C), and the corrected statement is that segmentation-guided classification reaches internal parity at best and never advantage, while the plain classifier remains ahead externally. We found this while auditing our own manuscript against the released raw data, after Section 4.13 had already been written to condemn exactly this error. The two validation rankings and the regime C result are pinned by the verification script so the selection cannot be lost again.

We are aware that this paper offers no new architecture, and we do not claim one. We submit it to MELBA because your stated scope includes empirical comparisons, because your reviewers are asked explicitly to assess reproducibility, and because this readership is the one that can act on the result.

Every number in the paper can be recomputed. The repository at https://github.com/Just-Ryan/burn-severity-benchmark contains the code, the trained weights, the split definition, the raw per-image predictions behind every table and figure, and `verify_paper_numbers.py`, a self-contained script that recomputes **220 published quantities** from the committed raw data without a GPU, a dataset download, or model weights; all 220 pass. Two of those checks exist specifically to catch us overstating our own results.

Use of generative AI. As required by MELBA's policy, we state how GenAI was used. The authors used Anthropic Claude (Claude Code; Opus 4.8 and Opus 5 models) to organise the project materials, to reproduce and check the reported results, to run the statistical, benchmarking and external-validation analyses — including the leak-free re-runs, the ten-seed replication, the perceptual-hash leakage check and the skin-tone probe — and to draft and edit the manuscript text. All drafting was performed from author-supplied results and under author direction; the authors verified every quantitative claim against the underlying result files, and the verification script named above exists so that a reviewer can do the same. No text was accepted without author review, and the tool was not used for unsupervised, de novo content creation. The research itself — the system design, model training, dataset preparation and the mobile application — was conceived and carried out by the authors, who take full responsibility for the content. This disclosure also appears in the Acknowledgements section of the manuscript. We would rather state this fully and let you judge it against your policy than describe it narrowly.

This manuscript is not under consideration elsewhere. An earlier and substantially shorter version was submitted to the Turkish Journal of Electrical Engineering and Computer Sciences and was declined at editorial screening, without external review, on 4 August 2026; we mention it for completeness and are happy to supply the correspondence. The present version adds the ten-seed replication, the 120-subset analysis, the skin-tone probe, the matched-architecture leakage contrast, the source-clustered intervals and the segmentation-split audit, none of which were in that submission. The work has not been published previously. All three authors have approved this submission, declare no conflicts of interest, and have not recently collaborated with any member of the MELBA editorial board.

Thank you for your consideration.

Sincerely,

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